

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459522.7119 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1792 +/- 0.0081
Transit depth (Rp/Rs)^2	3.21 +/- 0.29 [%]
Orbital Inclination (inc)	87.87 +/- 0.66
Airmass coefficient 1 (a1)	0.967 +/- 0.013
Airmass coefficient 2 (a2)	0.032 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.35 %
Best Comparison Star	#1 - [171, 178]
Optimal Aperture	3.96
Optimal Annulus	10.84
Transit Duration (day)	0.09 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1792 ± 0.0081 vs 0.1592 (deviation 1.93σ)
Duration fit vs published	0.09 d vs 0.0946 d (deviation 0.83σ)
Mid-time O–C	-2.6 min
Depth SNR	17.3
Residual scatter	1.35 %

Light curve

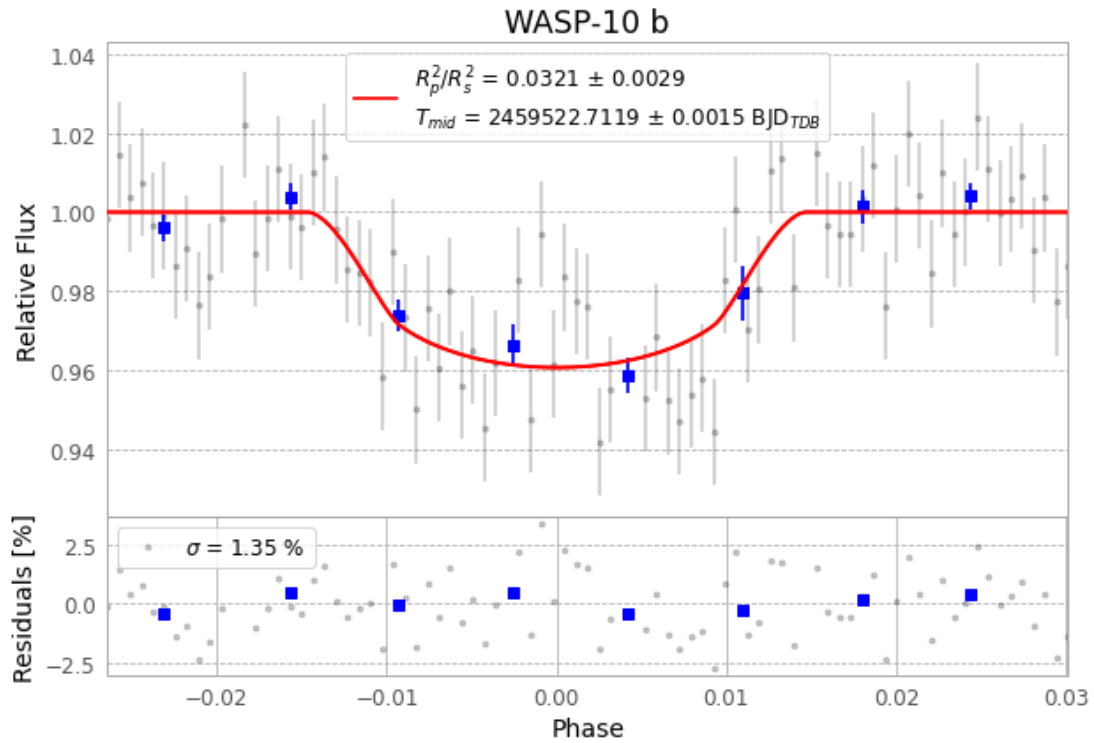


Figure 1 — FinalLightCurve_WASP-10 b_2021-11-03.png (SHA-256 170d8c88738cad34...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [473, 230], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dc80c18992830ead...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-10211104030013.FITS → WASP-10211104071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-211104.log (dc1012e5f5e3...), FinalLightCurve_WASP-10 b_2021-11-03.png (170d8c88738c...), FinalLightCurve_WASP-10 b_2021-11-03.csv (c67e6f2501bd...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:33Z.